

Water-Point Monitor

IoT-Based Smart Village Water Point Uptime Monitoring
System

Problem & Solution

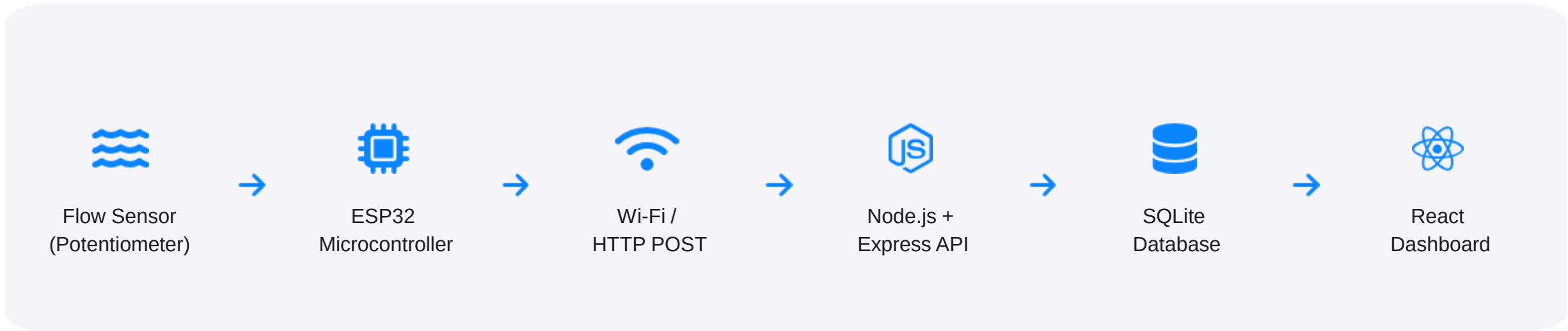
The Problem

- Manual monitoring of village water points.
- Delayed fault reporting.
- Significant water wastage.
- Long maintenance times.
- No real-time monitoring available.

Our Solution

WaterPoint Monitor uses an ESP32-based IoT device to monitor water flow, detect faults automatically, send data to a centralized server, and display live information on a smart dashboard.

System Architecture



Technologies Used



Frontend

[React.js](#) • [Bootstrap](#) • [Chart.js](#)

Dynamic, component-based user interface for real-time visualization.



Backend

[Node.js](#) • [Express.js](#)

Fast, scalable server architecture with robust RESTful API routing.



Database

[SQLite](#)

Lightweight, fast, relational data storage for point tracking and logs.

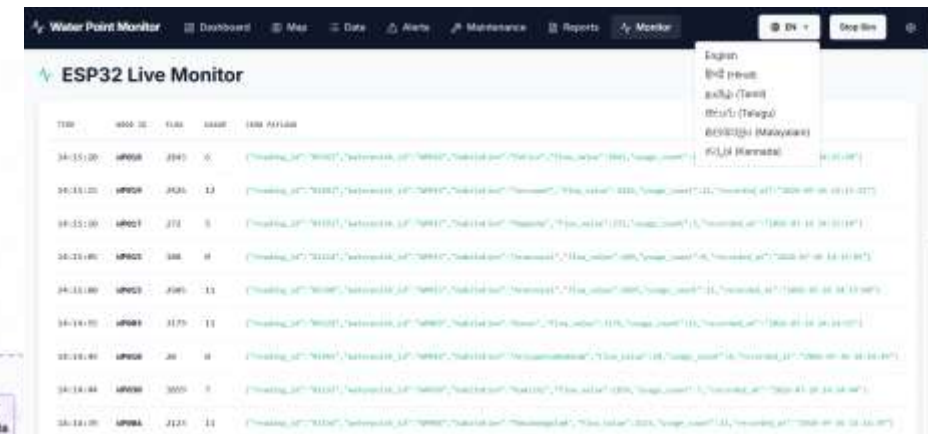
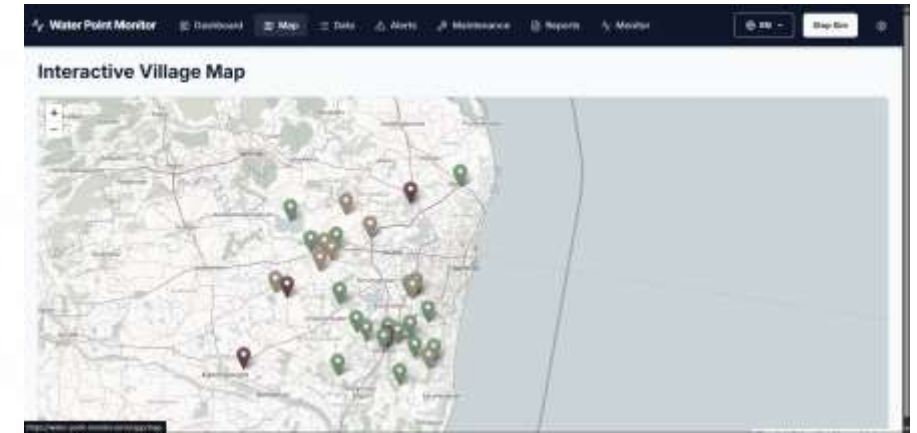
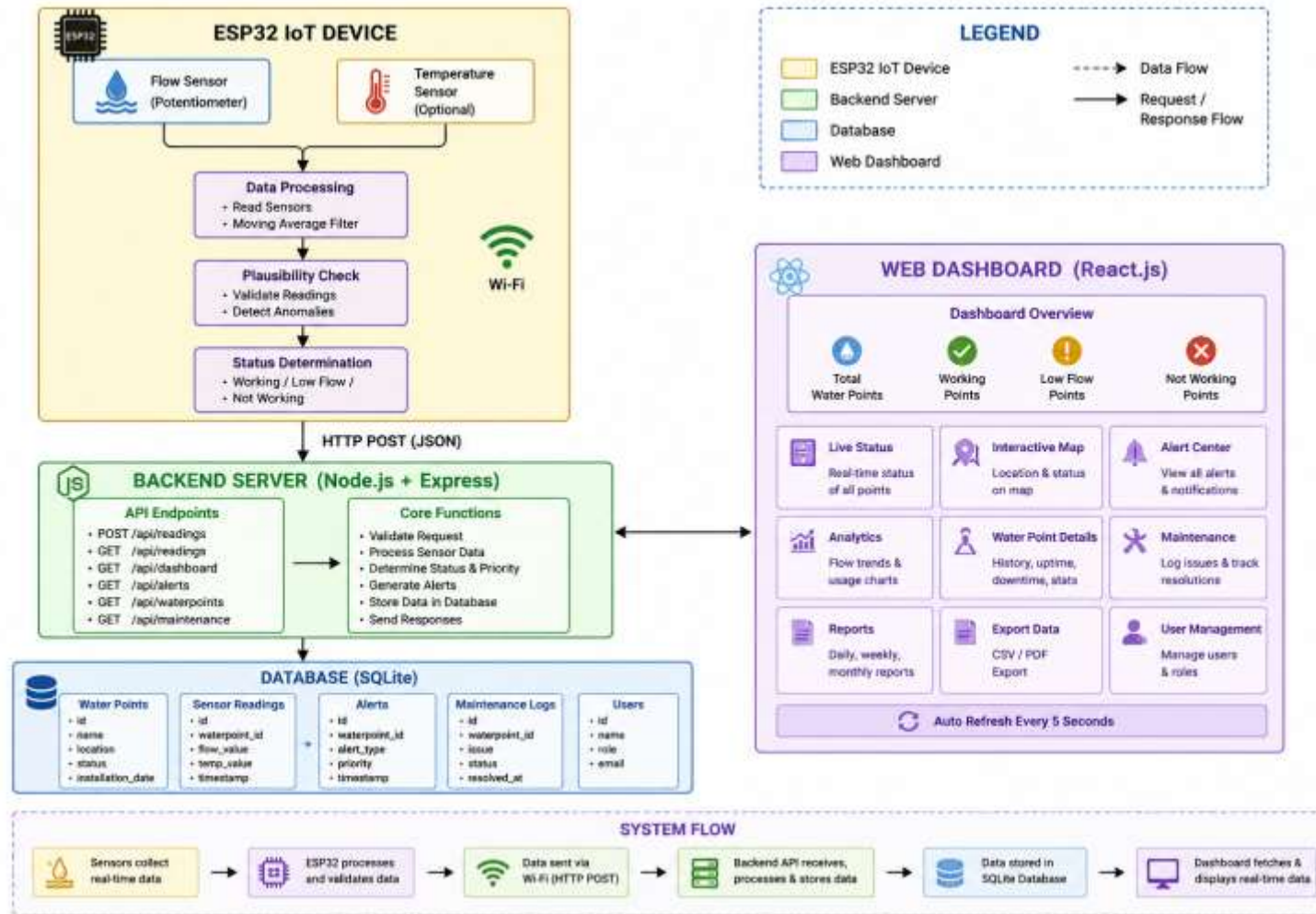


IoT & Hardware

[ESP32](#) • [Wokwi](#) • [Arduino IDE](#) • [REST API](#)

Low-cost edge processing, circuit simulation, and direct cloud communication.

Working Process



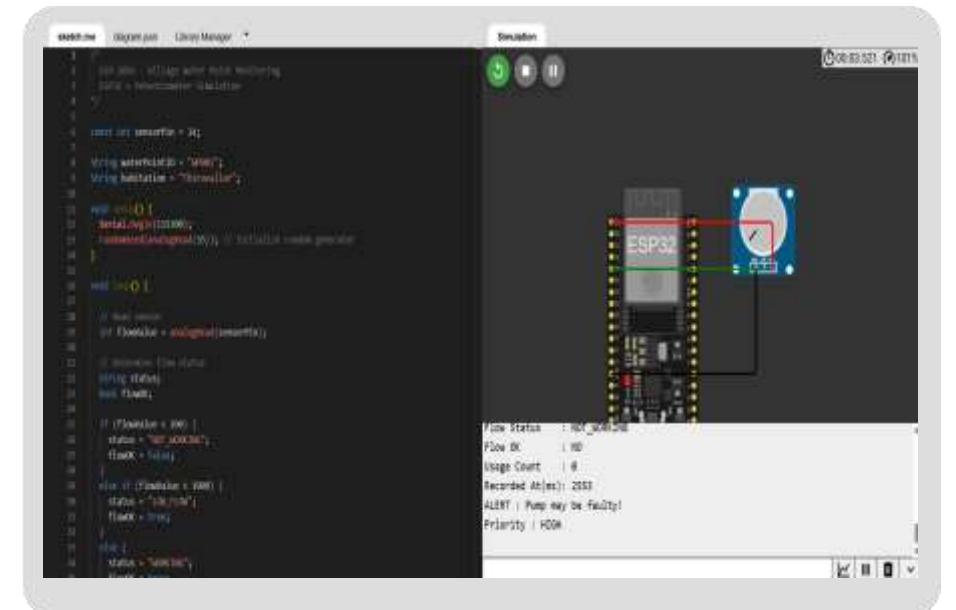
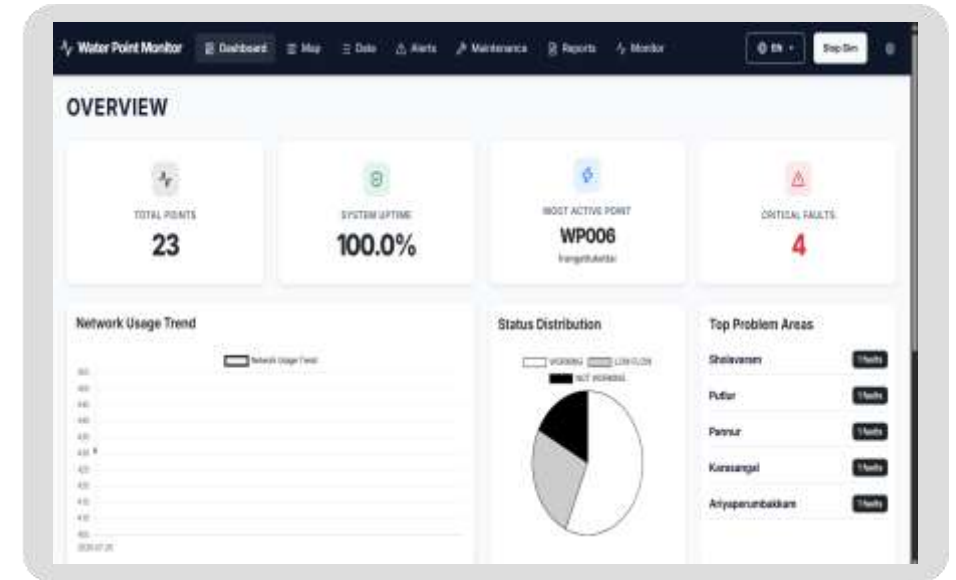
[Working Video DEMO LINK](#)

[Deployment Link](#)

Features & Demo

Dashboard Capabilities

- Live Water Point Status Mapping
- Real-time Flow Value Tracking
- Auto-categorization: Working / Low Flow / Not Working
- Historical Analytics Charts
- Instant Alert Notifications
- Digital Maintenance Logs



Advantages & Future Scope

Advantages

- Real-time Monitoring
- Automatic Fault Detection
- Low Cost & IoT-Based
- Easy Maintenance Routing
- Highly Scalable Architecture
- Village Friendly & Accessible

Future Scope

- Real Physical Flow Sensors
- GSM/4G & LoRa/NB-IoT Connectivity
- Dedicated Mobile Application
- SMS Alerts for Offline Areas
- AI-based Predictive Maintenance
- Full Cloud Deployment

Thank You

Questions?

 github.com/antony-jude/WaterPoint-Monitor